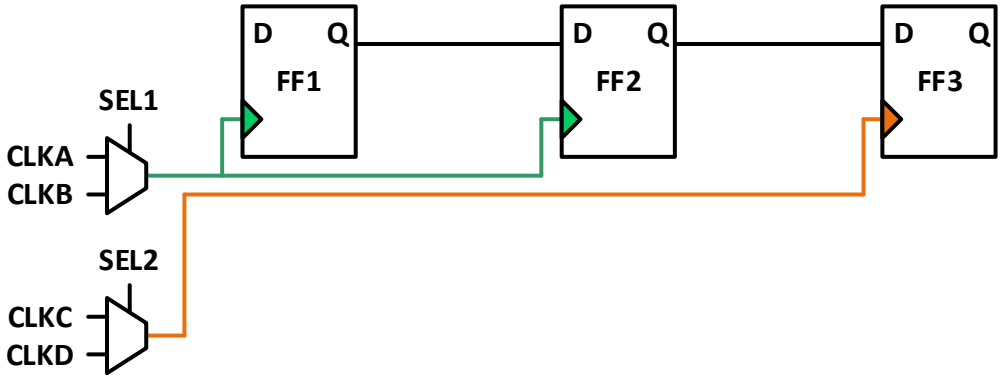
Advanced Computer-Aided VLSI System Design

Lab2: CDC & PLL

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Problem 1

In this problem, you will practice writing the synthesis constraints for the following CDC example.



Please run the Design Compiler synthesis with the appropriate constraints.

(1) Before adding the constraints for the CDC, how many timing paths from FF2/Q to FF3/D does Design Compiler check?

(Hint: report_timing -delay max -to FF3_reg/D -max_paths 10 -nworst 10)

(2) Take a screenshot of the timing paths from FF1/Q to FF2/D that Design Compiler checks after adding the constraints for the CDC.

(3) Take a screenshot of the timing paths from FF2/Q to FF3/D that Design Compiler checks after adding the constraints for the CDC.

Example of screenshot

```
Operating Conditions: slow   Library: slow
Wire Load Model Mode: top

  Startpoint: FF2_reg (rising edge-triggered flip-flop clocked by clk1)
  Endpoint: FF3_reg (rising edge-triggered flip-flop clocked by clk3)
  Path Group: (none)
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  cdc_example2       tsmc13_wl10           slow

  Point                                    Incr       Path
  -----------------------------------------------------------
  FF2_reg/CK (DFFQX1)                      0.00       0.00 r
  FF2_reg/Q (DFFQX1)                       0.40       0.40 r
  FF3_reg/D (DFFQX1)                       0.00       0.40 r
  data arrival time                                   0.40
  -----------------------------------------------------------
  (Path is unconstrained)
                              .
                              .
                              .
```

Problem 2

In this problem, you will learn the operation of the ADFP PLL. The following table lists the I/O signals of the PLL.

ADFP PLL

Path: /share1/tech/ADFP/Executable_Package/Collaterals/IP/pll/N16ADFP_PLL/VERILOG/n16adfp_pll_100a.v

Module name: N16ADFP_PLL

Signal	I/O Type	Description
FIN	Input	Input clock
[4:0] R	Input	Input clock divider
[7:0] F	Input	Frequency multiplier
[2:0] OD	Input	Output clock divider
PD	Input	Power down
BP	Input	Bypass
LD	Output	Lock
FOUT	Output	Output clock

Other signals: zeros.

```
N16ADFP_PLL U0(
        .FIN(clk_i),
        .VOPT0(1'b0), .VOPT1(1'b0), .VOPT2(1'b0),
        .RCON_S0(1'b0), .RCON_S1(1'b0), .RCON_S2(1'b0),
        .N25POFF(1'b0), .N25PON(1'b0), .N50PON(1'b0),
        .R0(r0), .R1(r1), .R2(r2), .R3(r3), .R4(r4),
        .F0(f0), .F1(f1), .F2(f2), .F3(f3), .F4(f4), .F5(f5), .F6(f6), .F7(f7),
        .PD(pd),
        .BP(1'b0),
        .OD0(od0), .OD1(od1), .OD2(od2),
        .LD(ld),
        .FOUT(clk_o)
);
```

The PLL output frequency

$$f_{out} = \left(\frac{f_{in}}{N_R}\right) \times N_F \times \frac{1}{N_{OD}} ,$$

where

$$N_R = R + 1, N_F = 2(F + 1), N_{OD} = \min\{2^{OD}, 2^4\}.$$

The PLL operation

Initialization: Set PD = 1 for 1000 ns.

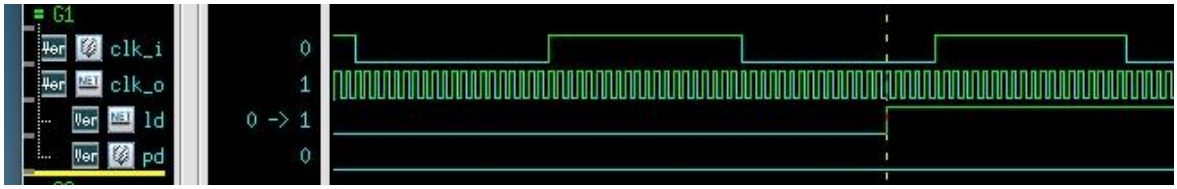
Locking: Set PD = 0 and BP = 0.

Some limits:

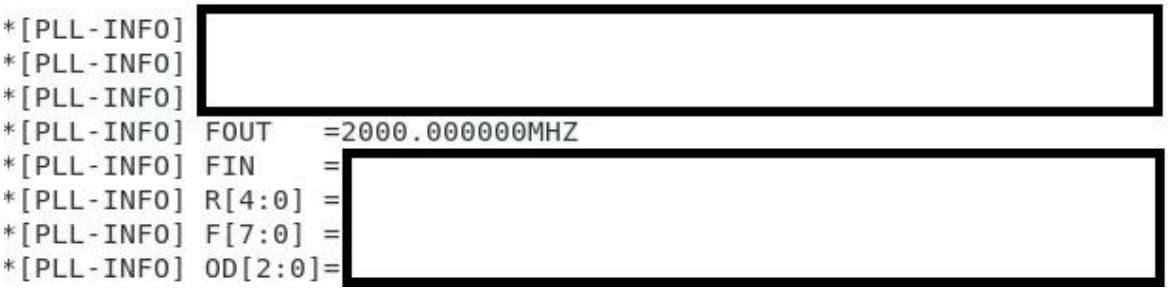
$$10\ MHz \leq f_{in} \leq 200\ MHz,$$
$$9.91\ MHz \leq f_{div} \leq 50.45\ MHz,$$
$$1486.5\ MHz \leq f_{vco} \leq 3027\ MHz.$$

Please run an RTL simulation of the ADFP PLL. The output frequency of the PLL should be 2000 MHz.

(1) Take a screenshot of the waveform that includes the signals FIN, LD = 1, and FOUT.



(2) Take a screenshot of the PLL information.



Submission

1. Due: March 24, 20:00.

2. The answer to Problem 1 (1) and the screenshots for Problem 1 (2) and (3).

3. The screenshots for Problem 2 (1) and (2).